

CLOUD SECURITY PBL

UNIT 1 — Evidence & Explanation Report

Responsibility Mapping & Asset Definition

Student Details

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Included Evidence

- Three-tier Web / App / DB network architecture
- VPC, subnets, Internet Gateway and routing
- Web, App and DB security groups
- S3 public-access blocking and encryption
- Terraform deployment, EC2, Session Manager and web-server verification

Submission Status

Infrastructure evidence is documented in the following pages.

SSH observation logs are included only as far as the supplied evidence supports; a full 24–48 hour observation must be completed before that requirement can be marked complete.

UNIT 1 — Evidence Summary

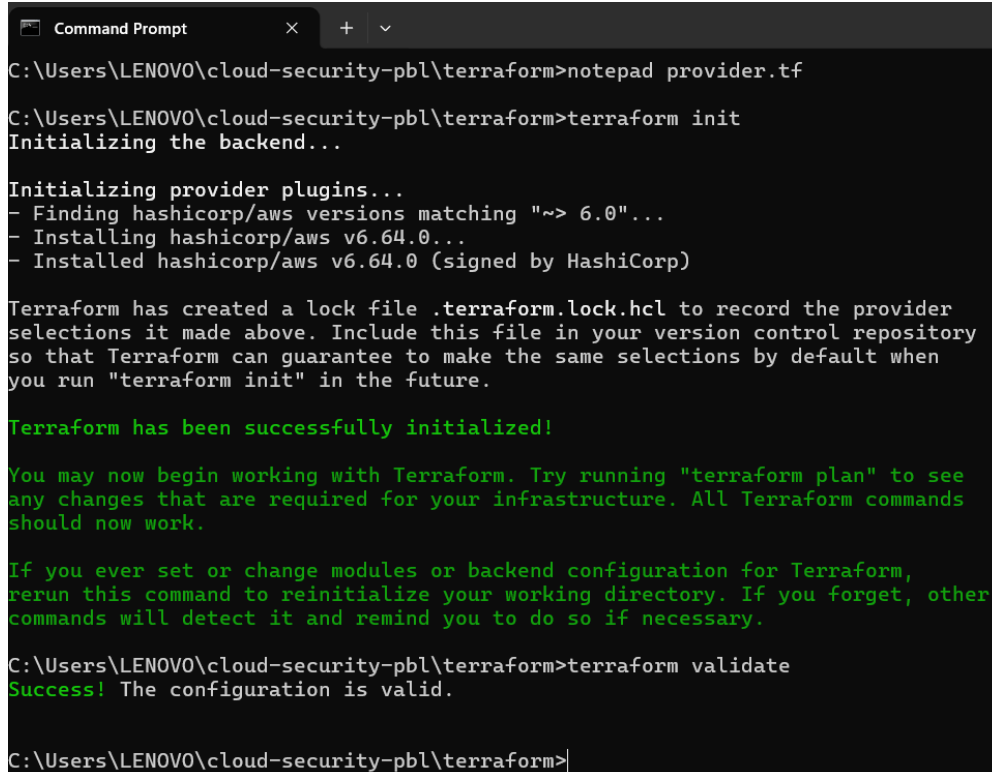
No.	Evidence	Purpose / Section	Status
1	Terraform Initialization and Validation	Unit 1 → Terraform / Infrastructure Preparation	Current evidence
2	Three-Tier Subnet and VPC Definition	Unit 1 → 3-Tier Network Architecture	Current evidence
3	Web Tier Security Group	Unit 1 → Network Security / Security Groups	Current evidence
4	Public Route Table and Routing	Unit 1 → Routing / Internet Connectivity	Current evidence
5	Internet Gateway and Infrastructure	Unit 1 → VPC Connectivity	Current evidence
6	Public Route Table — Duplicate Evidence	Unit 1 → Routing	Duplicate/reference
7	Internet Gateway — Duplicate Evidence	Unit 1 → VPC Connectivity	Duplicate/reference
8	Public Route Table — Reference Copy	Unit 1 → Routing	Duplicate/reference
9	S3 Public Access Block and Encryption	Unit 1 → Secure Storage	Current evidence
10	S3 Security Configuration — Duplicate Evidence	Unit 1 → Secure Storage	Duplicate/reference
11	Database Tier Security Group	Unit 1 → Network Security / Database Tier	Current evidence
12	Application Tier Security Group	Unit 1 → Network Security / Application Tier	Current evidence
13	Terraform Deployment — 29 Resources	Unit 1 → Terraform Deployment	Current evidence
14	Current EC2 Instance	Unit 1 → EC2 Deployment	Current evidence
15	Systems Manager Session Manager Target	Unit 1 → Secure EC2 Administration	Current evidence
16	SSM, Nginx and Local Web Server Verification	Unit 1 → Web Server Verification	Current evidence
17	Current Web Server Browser Test	Unit 1 → Web Server Verification	Current evidence
18	Initial Observation EC2 Instance	Unit 1 → Internet-Facing Observation / Historical Evidence	Historical — do not treat as current
19	Initial Observation Web Server	Unit 1 → Internet-Facing Observation / Historical Evidence	Historical — do not treat as current
20	EC2 Replacement Plan	Unit 1 → Deployment Change / Historical Evidence	Historical/change evidence

Evidence 1: Terraform Initialization and Validation

Place in: Unit 1 → Terraform / Infrastructure Preparation

Status: Current evidence

What it proves: Shows successful Terraform initialization and `terraform validate` returning that the configuration is valid.



```
Command Prompt
C:\Users\LENOVO\cloud-security-pbl\terraform>notepad provider.tf

C:\Users\LENOVO\cloud-security-pbl\terraform>terraform init
Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 6.0"...
- Installing hashicorp/aws v6.64.0...
- Installed hashicorp/aws v6.64.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

C:\Users\LENOVO\cloud-security-pbl\terraform>terraform validate
Success! The configuration is valid.

C:\Users\LENOVO\cloud-security-pbl\terraform>
```

Figure 1: Terraform Initialization and Validation.

Evidence 2: Three-Tier Subnet and VPC Definition

Place in: Unit 1 → 3-Tier Network Architecture

Status: Current evidence

What it proves: Shows Web, App and DB subnet resources and the VPC CIDR. This documents the logical three-tier network layout.

```
Command Prompt
}
# aws_subnet_db_subnet will be created
resource "aws_subnet" "db_subnet" {
  arm = (Unknown after apply)
  assign_ipv6_address_on_creation = false
  availability_zone = "ap-south-1b"
  cidr_block = "10.0.1.0/24"
  enable_dns64 = false
  enable_resource_name_dns_a_record_on_launch = false
  enable_resource_name_dns_aaaa_record_on_launch = false
  id = (Unknown after apply)
  ipv6_cidr_block = (Unknown after apply)
  ipv6_cidr_block_association_id = (Unknown after apply)
  ipv6_native = false
  map_public_ip_on_launch = false
  owner_id = (Unknown after apply)
  private_dns_hostnames_type_on_launch = (Unknown after apply)
  region = "ap-south-1"
  tags = {
    "Name" = "PBL-DB-Private-Subnet"
    "Tier" = "Database"
  }
  tags_all = {
    "Name" = "PBL-DB-Private-Subnet"
    "Tier" = "Database"
  }
  vpc_id = (Unknown after apply)
}

# aws_subnet_web_subnet will be created
resource "aws_subnet" "web_subnet" {
  arm = (Unknown after apply)
  assign_ipv6_address_on_creation = false
  availability_zone = "ap-south-1a"
  cidr_block = "10.0.1.0/24"
  enable_dns64 = false
  enable_resource_name_dns_a_record_on_launch = false
  enable_resource_name_dns_aaaa_record_on_launch = false
  id = (Unknown after apply)
  ipv6_cidr_block = (Unknown after apply)
  ipv6_cidr_block_association_id = (Unknown after apply)
  ipv6_native = false
  map_public_ip_on_launch = true
  owner_id = (Unknown after apply)
  private_dns_hostnames_type_on_launch = (Unknown after apply)
  region = "ap-south-1"
  tags = {
    "Name" = "PBL-Web-Public-Subnet"
    "Tier" = "Web"
  }
  tags_all = {
    "Name" = "PBL-Web-Public-Subnet"
    "Tier" = "Web"
  }
  vpc_id = (Unknown after apply)
}

# aws_vpc.pbl_vpc will be created
resource "aws_vpc" "pbl_vpc" {
  arm = (Unknown after apply)
  cidr_block = "10.0.0.0/16"
  default_network_acl_id = (Unknown after apply)
  default_route_table_id = (Unknown after apply)
  default_security_group_id = (Unknown after apply)
  dhcp_options_id = (Unknown after apply)
  enable_dns_hostnames = true
  enable_dns_support = true
  enable_network_address_usage_metrics = (Unknown after apply)
  id = (Unknown after apply)
  instance_tenancy = "default"
  ipv6_association_id = (Unknown after apply)
  ipv6_cidr_block = (Unknown after apply)
  ipv6_cidr_block_network_border_group = (Unknown after apply)
  main_route_table_id = (Unknown after apply)
  owner_id = (Unknown after apply)
  region = "ap-south-1"
  tags = {
    "Environment" = "pbl"
    "Name" = "PBL-vpc"
    "Project" = "Cloud-Security-PBL"
  }
}
```

Figure 2: Three-Tier Subnet and VPC Definition.

Evidence 3: Web Tier Security Group

Place in: Unit 1 → Network Security / Security Groups

Status: Current evidence

What it proves: Shows the Web security group with HTTP 80 and HTTPS 443 permitted from the Internet.

```
Command Prompt

# aws_security_group.web_sg will be created
resource "aws_security_group" "web_sg" {
  name               = "web_sg"
  description        = "(unknown after apply)"
  vpc_id             = "vpc-0a1b2c3d"
  egress              = [
    {
      cidr_blocks      = [
        "0.0.0.0/0"
      ]
      description      = "Allow outbound traffic"
      from_port         = 0
      ipv6_cidr_blocks = []
      prefix_list_ids   = []
      protocol          = "all"
      security_groups   = []
      self              = false
      to_port           = 0
    }
  ]
  id                 = "(unknown after apply)"
  ingress             = [
    {
      cidr_blocks      = [
        "0.0.0.0/0"
      ]
      description      = "HTTP from Internet"
      from_port         = 80
      ipv6_cidr_blocks = []
      prefix_list_ids   = []
      protocol          = "tcp"
      security_groups   = []
      self              = false
      to_port           = 80
    },
    {
      cidr_blocks      = [
        "0.0.0.0/0"
      ]
      description      = "HTTPS from Internet"
      from_port         = 443
      ipv6_cidr_blocks = []
      prefix_list_ids   = []
      protocol          = "tcp"
      security_groups   = []
      self              = false
      to_port           = 443
    }
  ]
  name               = "PBL-Web-SG"
  name_prefix        = "(unknown after apply)"
  owner_id           = "(unknown after apply)"
  region             = "us-east-1"
  revoke_rules_on_delete = false
  tags               = [
    {
      "Name" = "PBL-Web-SG"
      "Project" = "Cloud-Security-PBL"
      "Tier" = "Web"
    }
  ]
  tags_all           = [
    {
      "Name" = "PBL-Web-SG"
      "Project" = "Cloud-Security-PBL"
      "Tier" = "Web"
    }
  ]
  vpc_id             = "(unknown after apply)"
}

# aws_subnet.app_subnet will be created
resource "aws_subnet" "app_subnet" {
  vpc_id              = "vpc-0a1b2c3d"
  availability_zone    = "us-east-1a"
  cidr_block           = "10.0.2.0/24"
  enable_dns           = false
  enable_dns_hostnames = false
  enable_resource_name_dns_record_on_launch = false
  id                  = "(unknown after apply)"
  ipv6_cidr_block      = "(unknown after apply)"
  ipv6_cidr_block_association_id = "(unknown after apply)"
  ipv6_only            = false
  map_public_ip_on_launch = false
  owner_id             = "(unknown after apply)"
  private_dns_hostnames_type_on_launch = "(unknown after apply)"
}
```

Figure 3: Web Tier Security Group.

Evidence 4: Public Route Table and Routing

Place in: Unit 1 → Routing / Internet Connectivity

Status: Current evidence

What it proves: Shows the public route table with a default route to the Internet Gateway and related route-table associations.

```
Command Prompt

# aws_route_table-public-rt will be created
+ resource "aws_route_table" "public-rt" {
+   arn = (known after apply)
+   id = (known after apply)
+   owner_id = (known after apply)
+   propagating_vpcs = (known after apply)
+   region = "ap-south-1"
+   route = [
+     {
+       cidr_block = "0.0.0.0/0"
+       gateway_id = (known after apply)
+       # (12 unchanged attributes hidden)
+     }
+   ]
+   tags = {
+     "Name" = "PMU-Public-Route-Table"
+   }
+   tags_all = {
+     "Name" = "PMU-Public-Route-Table"
+   }
+   vpc_id = (known after apply)
+ }

# aws_route_table_association-app-private-association will be created
+ resource "aws_route_table_association" "app-private-association" {
+   id = (known after apply)
+   region = "ap-south-1"
+   route_table_id = (known after apply)
+   subnet_id = (known after apply)
+ }

# aws_route_table_association-db-private-association will be created
+ resource "aws_route_table_association" "db-private-association" {
+   id = (known after apply)
+   region = "ap-south-1"
+   route_table_id = (known after apply)
+   subnet_id = (known after apply)
+ }

# aws_route_table_association-web-public-association will be created
+ resource "aws_route_table_association" "web-public-association" {
+   id = (known after apply)
+   region = "ap-south-1"
+   route_table_id = (known after apply)
+   subnet_id = (known after apply)
+ }

# aws_s3_bucket-pbl-bucket will be created
+ resource "aws_s3_bucket" "pbl-bucket" {
+   acceleration_status = (known after apply)
+   acl = (known after apply)
+   arn = (known after apply)
+   bucket = (known after apply)
+   bucket_domain_name = (known after apply)
+   bucket_encryption = (known after apply)
+   bucket_prefix = "pbl-cloud-security-"
+   bucket_region = (known after apply)
+   bucket_regional_domain_name = (known after apply)
+   force_destroy = false
+   hosted_zone_id = (known after apply)
+   id = (known after apply)
+   object_lock_enabled = (known after apply)
+   policy = (known after apply)
+   region = "ap-south-1"
+   request_payer = (known after apply)
+   tags = [
+     {
+       "Name" = "PMU-S3-Bucket"
+       "Project" = "Cloud-Security-PMU"
+     }
+   ]
+   tags_all = {
+     "Name" = "PMU-S3-Bucket"
+     "Project" = "Cloud-Security-PMU"
+   }
+   website_domain = (known after apply)
+   website_endpoint = (known after apply)
+   cors_rule (known after apply)
+   grant (known after apply)
+   lifecycle_rule (known after apply)
+   logging (known after apply)
+ }
```

Figure 4: Public Route Table and Routing.

Evidence 5: Internet Gateway and Infrastructure

Place in: Unit 1 → VPC Connectivity

Status: Current evidence

What it proves: Shows the Internet Gateway and related Terraform resources used by the VPC.

```
Command Prompt

<title>Cloud Security PBL</title>
</head>
<body>
  <div>Cloud Security PBL Web Server</div>
  <p>Web server is running successfully.</p>
</body>
</html>

systemctl restart nginx

echo "PBL observation server initialized at $(date)" \
  >> /usr/local/bin/observation.log

ENV
+ user_data_bas64 = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids = (known after apply)
+ capacity_reservation_specification (known after apply)
+ cpu_options (known after apply)
+ ebs_block_device (known after apply)
+ enclave_options (known after apply)
+ ephemeral_block_device (known after apply)
+ instance_market_options (known after apply)
+ maintenance_options (known after apply)
+ metadata_options (known after apply)
+ network_interface (known after apply)
+ primary_network_interface (known after apply)
+ private_dns_name_options (known after apply)
+ root_block_device {
+   delete_on_termination = true
+   device_name = (known after apply)
+   encrypted = true
+   iops = (known after apply)
+   kms_key_id = (known after apply)
+   tags_all = (known after apply)
+   throughput = (known after apply)
+   volume_id = (known after apply)
+   volume_size = 8
+   volume_type = "gp3"
+ }
+ secondary_network_interface (known after apply)
}

# aws_internet_gateway.pbl_igw will be created
resource "aws_internet_gateway" "pbl_igw" {
  az = (known after apply)
  id = (known after apply)
  owner_id = (known after apply)
  region = "ap-south-1"
  tags = [
    "Name" = "PBL-Internet-Gateway",
    "Project" = "Cloud-Security-PBL"
  ]
  tags_all = [
    "Name" = "PBL-Internet-Gateway",
    "Project" = "Cloud-Security-PBL"
  ]
  vpc_id = (known after apply)
}

# aws_route_table.private_rt will be created
resource "aws_route_table" "private_rt" {
  az = (known after apply)
  id = (known after apply)
  owner_id = (known after apply)
  propagating_vpcs = (known after apply)
  region = "ap-south-1"
  route = (known after apply)
  tags = [
    "Name" = "PBL-Private-Route-Table"
  ]
}
```

Figure 5: Internet Gateway and Infrastructure.

Evidence 7: Internet Gateway — Duplicate Evidence

Place in: Unit 1 → VPC Connectivity

Status: Duplicate/reference

What it proves: Duplicate copy of the Internet Gateway evidence supplied in the screenshots.

```

Command Prompt
<title>Cloud Security PBL</title>
</head>
<body>
<div>Cloud Security PBL Web Server</div>
<p>Web server is running successfully.</p>
</body>
</html>
HTML

systemctl restart nginx

echo "PBL observation server initialized at $(date)" \
  > /var/log/pbl-observation.log

EOF
+ user_data_b64encode = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids = (known after apply)
+ capacity_reservation_specification (known after apply)
+ cpu_options (known after apply)
+ ebs_block_device (known after apply)
+ enclave_options (known after apply)
+ ephemeral_block_device (known after apply)
+ instance_market_options (known after apply)
+ maintenance_options (known after apply)
+ metadata_options (known after apply)
+ network_interface (known after apply)
+ primary_network_interface (known after apply)
+ private_dns_name_options (known after apply)
+ root_block_device {
+   delete_on_termination = true
+   device_name = (known after apply)
+   encrypted = true
+   iops = (known after apply)
+   new_key_id = (known after apply)
+   tags_all = (known after apply)
+   throughput = (known after apply)
+   volume_id = (known after apply)
+   volume_size = 8
+   volume_type = "gp3"
+ }
+ secondary_network_interface (known after apply)
}

# aws_internet_gateway.pbl_igw will be created
resource "aws_internet_gateway" "pbl_igw" {
+ arn = (known after apply)
+ id = (known after apply)
+ owner_id = (known after apply)
+ region = "ap-south-1"
+ tags = [
+   "Name" = "PBL-Internet-Gateway"
+   "Project" = "Cloud-Security-PBL"
+ ]
+ tags_all = {
+   "Name" = "PBL-Internet-Gateway"
+   "Project" = "Cloud-Security-PBL"
+ }
+ vpc_id = (known after apply)
}

# aws_route_table.private_rt will be created
resource "aws_route_table" "private_rt" {
+ arn = (known after apply)
+ id = (known after apply)
+ owner_id = (known after apply)
+ propagating_vpcs = (known after apply)
+ region = "ap-south-1"
+ route = (known after apply)
+ tags = [
+   "Name" = "PBL-Private-Route-Table"
+ ]
}

```

Figure 7: Internet Gateway — Duplicate Evidence.

Evidence 9: S3 Public Access Block and Encryption

Place in: Unit 1 → Secure Storage

Status: Current evidence

What it proves: Shows S3 public-access blocking and AES256 server-side encryption configuration.

```
Command Prompt

# aws_s3_bucket_public_access_block.pbl_bucket_block will be created
resource "aws_s3_bucket_public_access_block" "pbl_bucket_block" {
  bucket_public_access = true
  bucket_public_policy = true
  bucket = (known after apply)
  id = (known after apply)
  ignore_public_access = true
  region = "ap-south-1"
  restrict_public_buckets = true
}

# aws_s3_bucket_server_side_encryption_configuration.pbl_bucket_encryption will be created
resource "aws_s3_bucket_server_side_encryption_configuration" "pbl_bucket_encryption" {
  bucket = (known after apply)
  id = (known after apply)
  region = "ap-south-1"

  rule {
    blocked_encryption_types = (known after apply)
    bucket_key_enabled = (known after apply)
    apply_server_side_encryption_by_default {
      kms_master_key_id = (known after apply)
      sse_algorithm = "AES256"
    }
  }
}

# aws_security_group_app_sg will be created
resource "aws_security_group" "app_sg" {
  name = "Security group for the application tier"
  description = "Security group for the application tier"
  ingress = [
    {
      cidr_blocks = [
        "0.0.0.0/0"
      ]
      description = "Allow outbound traffic"
      from_port = 0
      ip_protocol = "all"
      prefix_list_ids = []
      protocol = "all"
      security_groups = []
      self = false
      to_port = 0
    },
  ],
  id = (known after apply)
  ingress = [
    {
      cidr_blocks = [
        "0.0.0.0/0"
      ]
      description = "Application traffic from Web Tier"
      from_port = 8080
      ip_protocol = "all"
      prefix_list_ids = []
      protocol = "tcp"
      security_groups = (known after apply)
      self = false
      to_port = 8080
    },
  ],
  name = "PBL-App-SG"
  name_prefix = (known after apply)
  owner_id = (known after apply)
  region = "ap-south-1"
  revoke_rules_on_delete = false
  tags = {
    "Name" = "PBL-App-SG"
    "Project" = "Cloud-Security-PBL"
    "Tier" = "App"
  }
  tags_all = [
    {
      "Name" = "PBL-App-SG"
      "Project" = "Cloud-Security-PBL"
      "Tier" = "App"
    }
  ],
  vpc_id = (known after apply)
}

# aws_security_group_db_sg will be created
resource "aws_security_group" "db_sg" {
  name = "Security group for the database tier"
  description = "Security group for the database tier"
  ingress = [
    {
      cidr_blocks = [
        "0.0.0.0/0"
      ]
      description = "Allow inbound traffic"
      from_port = 0
      ip_protocol = "all"
      prefix_list_ids = []
      protocol = "all"
      security_groups = []
      self = false
      to_port = 0
    },
  ],
  id = (known after apply)
  ingress = [
    {
      cidr_blocks = [
        "0.0.0.0/0"
      ]
      description = "Application traffic from Web Tier"
      from_port = 8080
      ip_protocol = "all"
      prefix_list_ids = []
      protocol = "tcp"
      security_groups = (known after apply)
      self = false
      to_port = 8080
    },
  ],
  name = "PBL-DB-SG"
  name_prefix = (known after apply)
  owner_id = (known after apply)
  region = "ap-south-1"
  revoke_rules_on_delete = false
  tags = {
    "Name" = "PBL-DB-SG"
    "Project" = "Cloud-Security-PBL"
    "Tier" = "DB"
  }
  tags_all = [
    {
      "Name" = "PBL-DB-SG"
      "Project" = "Cloud-Security-PBL"
      "Tier" = "DB"
    }
  ],
  vpc_id = (known after apply)
}
```

Figure 9: S3 Public Access Block and Encryption.

Evidence 11: Database Tier Security Group

Place in: Unit 1 → Network Security / Database Tier

Status: Current evidence

What it proves: Shows the database security group and PostgreSQL port 5432 restricted to application-tier traffic.

```
Command Prompt
# aws_security_group.db_sg will be created
resource "aws_security_group" "db_sg" {
  arn = (unknown after apply)
  description = "Security group for the database tier"
  egress = [
    {
      cidr_blocks = [
        "0.0.0.0/0",
      ]
      description = "Allow outbound traffic"
      from_port = 0
      ipv6_cidr_blocks = []
      prefix_list_ids = []
      protocol = "-1"
      security_groups = []
      self = false
      to_port = 0
    }
  ],
  id = (unknown after apply)
  ingress = [
    {
      cidr_blocks = []
      description = "PostgreSQL from App Tier"
      from_port = 5432
      ipv6_cidr_blocks = []
      prefix_list_ids = []
      protocol = "tcp"
      security_groups = (unknown after apply)
      self = false
      to_port = 5432
    }
  ],
  name = "PBL-DB-SG"
  name_prefix = (unknown after apply)
  owner_id = (unknown after apply)
  region = "ap-south-1"
  revoke_rules_on_delete = false
  tags = {
    "Name" = "PBL-DB-SG"
    "Project" = "Cloud-Security-PBL"
    "Tier" = "Database"
  }
  tags_all = {
    "Name" = "PBL-DB-SG"
    "Project" = "Cloud-Security-PBL"
    "Tier" = "Database"
  }
  vpc_id = (unknown after apply)
}

# aws_security_group.observation_sg will be created
resource "aws_security_group" "observation_sg" {
  arn = (unknown after apply)
  description = "Temporary SG for passive SSH threat observation"
  egress = [
    {
      cidr_blocks = [
        "0.0.0.0/0",
      ]
      description = "Allow outbound traffic"
      from_port = 0
      ipv6_cidr_blocks = []
      prefix_list_ids = []
      protocol = "-1"
      security_groups = []
      self = false
      to_port = 0
    }
  ],
  id = (unknown after apply)
  ingress = [
    {
      cidr_blocks = [
        "0.0.0.0/0",
      ]
      description = "HTTP observation - Internet"
      from_port = 80
      ipv6_cidr_blocks = []
      prefix_list_ids = []
      protocol = "tcp"
      security_groups = []
      self = false
      to_port = 80
    }
  ],
  name = "PBL-OBS-SG"
  name_prefix = (unknown after apply)
  owner_id = (unknown after apply)
  region = "ap-south-1"
  revoke_rules_on_delete = false
  tags = {
    "Name" = "PBL-OBS-SG"
    "Project" = "Cloud-Security-PBL"
    "Tier" = "Observation"
  }
  tags_all = {
    "Name" = "PBL-OBS-SG"
    "Project" = "Cloud-Security-PBL"
    "Tier" = "Observation"
  }
  vpc_id = (unknown after apply)
}
```

Figure 11: Database Tier Security Group.

Evidence 12: Application Tier Security Group

Place in: Unit 1 → Network Security / Application Tier

Status: Current evidence

What it proves: Shows the application-tier security group and TCP port 8080 for traffic from the Web tier.

```
Command Prompt

# aws_route_table.public_rt will be created
resource "aws_route_table" "public_rt" {
  arn          = (known after apply)
  id           = (known after apply)
  owner_id     = (known after apply)
  propagating_vgms = (known after apply)
  region       = "ap-south-1"
  route       = [
    {
      cidr_block      = "0.0.0.0/0"
      gateway_id      = (known after apply)
    }
  ]
  tags         = {
    "Name" = "PBL-Public-Route-Table"
  }
  tags_all     = {
    "Name" = "PBL-Public-Route-Table"
  }
  vpc_id       = (known after apply)
}

# aws_route_table_association.app_private_association will be created
resource "aws_route_table_association" "app_private_association" {
  id           = (known after apply)
  region       = "ap-south-1"
  route_table_id = (known after apply)
  subnet_id    = (known after apply)
}

# aws_route_table_association.db_private_association will be created
resource "aws_route_table_association" "db_private_association" {
  id           = (known after apply)
  region       = "ap-south-1"
  route_table_id = (known after apply)
  subnet_id    = (known after apply)
}

# aws_route_table_association.web_public_association will be created
resource "aws_route_table_association" "web_public_association" {
  id           = (known after apply)
  region       = "ap-south-1"
  route_table_id = (known after apply)
  subnet_id    = (known after apply)
}

# aws_s3_bucket.pbl_bucket will be created
resource "aws_s3_bucket" "pbl_bucket" {
  acceleration_status = (known after apply)
  acl                 = (known after apply)
  arn                 = (known after apply)
  bucket             = (known after apply)
  bucket_domain_name = (known after apply)
  bucket_name        = (known after apply)
  bucket_prefix      = "pbl-cloud-security-"
  bucket_region      = (known after apply)
  bucket_regional_domain_name = (known after apply)
  force_destroy      = false
  hosted_zone_id     = (known after apply)
  id                 = (known after apply)
  object_lock_enabled = (known after apply)
  policy             = (known after apply)
  region             = "ap-south-1"
  request_payer      = (known after apply)
  tags               = {
    "Name" = "PBL-S3-Bucket"
    "Project" = "Cloud-Security-PBL"
  }
  tags_all           = {
    "Name" = "PBL-S3-Bucket"
    "Project" = "Cloud-Security-PBL"
  }
  website_domain     = (known after apply)
  website_endpoint   = (known after apply)
  cors_rule          = (known after apply)
  grant              = (known after apply)
  lifecycle_rule     = (known after apply)
  logging             = (known after apply)
}
```

Figure 12: Application Tier Security Group.

Evidence 13: Terraform Deployment — 29 Resources

Place in: Unit 1 → Terraform Deployment

Status: Current evidence

What it proves: Shows Terraform applying the PBL infrastructure and the successful creation process. The final output records 29 resources added.

```
Plan: 29 to add, 0 to change, 0 to destroy.

C:\Users\LENOVO\cloud-security-pbl\terraform>terraform apply pbl.tfplan
aws_iam_role.ec2_s3_role: Creating...
aws_vpc.pbl_vpc: Creating...
aws_iam_policy.auditor_policy: Creating...
aws_iam_group.developers: Creating...
aws_iam_group.auditors: Creating...
aws_s3_bucket.pbl_bucket: Creating...
aws_iam_group.auditors: Creation complete after 4s [id=PBL-Auditors]
aws_iam_group.developers: Creation complete after 4s [id=PBL-Developers]
aws_iam_policy.auditor_policy: Creation complete after 4s [id=arn:aws:iam::181137999524:policy/PBL-Auditor-ReadOnly]
aws_iam_group_policy_attachment.auditor_policy_attachment: Creating...
aws_iam_role.ec2_s3_role: Creation complete after 4s [id=PBL-EC2-S3-Role]
aws_iam_role_policy_attachment.ec2_s3_policy_attachment: Creating...
aws_iam_instance_profile.ec2_profile: Creating...
aws_iam_group_policy_attachment.auditor_policy_attachment: Creation complete after 1s [id=PBL-Auditors-382609161940358000000000]
aws_iam_role_policy_attachment.ec2_s3_policy_attachment: Creation complete after 1s [id=PBL-EC2-S3-Role/arn:aws:iam::aws:policy/AWSManagedInstanceCore]
aws_s3_bucket.pbl_bucket: Creation complete after 6s [id=pbl-cloud-security-371253588a3a74db423aadd4d3]
aws_s3_bucket_public_access_block.pbl_bucket_block: Creating...
aws_iam_policy.ec2_s3_policy: Creating...
aws_iam_policy.developer_policy: Creating...
aws_s3_bucket_server_side_encryption_configuration.pbl_bucket_encryption: Creating...
aws_vpc.pbl_vpc: Creation complete after 6s [id=vpc-0276b3f4HedeSle8a]
aws_route_table.private_rt: Creating...
aws_internet_gateway.pbl_igw: Creating...
aws_subnet.web_subnet: Creating...
aws_subnet.app_subnet: Creating...
aws_security_group.observation_sg: Creating...
aws_s3_bucket_public_access_block.pbl_bucket_block: Creation complete after 1s [id=pbl-cloud-security-371253588a3a74db423aadd4d3]
aws_subnet.db_subnet: Creating...
aws_iam_policy.ec2_s3_policy: Creation complete after 1s [id=arn:aws:iam::181137999524:policy/PBL-EC2-S3-LeastPrivilege]
aws_security_group.web_sg: Creating...
aws_s3_bucket_server_side_encryption_configuration.pbl_bucket_encryption: Creation complete after 1s [id=pbl-cloud-security-371253588a3a74db423aadd4d3]
aws_iam_role_policy_attachment.ec2_s3_policy_attachment: Creating...
aws_iam_policy.developer_policy: Creation complete after 1s [id=arn:aws:iam::181137999524:policy/PBL-Developer-LeastPrivilege]
aws_iam_group_policy_attachment.developer_policy_attachment: Creating...
aws_iam_role_policy_attachment.ec2_s3_policy_attachment: Creation complete after 1s [id=PBL-EC2-S3-Role/arn:aws:iam::181137999524:policy/PBL-EC2-S3-LeastPrivilege]
aws_iam_group_policy_attachment.developer_policy_attachment: Creation complete after 1s [id=PBL-Developers-2f2689181e485c7880808092]
aws_subnet.app_subnet: Creation complete after 4s [id=subnet-080dfc612968fa38b]
aws_internet_gateway.pbl_igw: Creation complete after 5s [id=igw-0f0dc9a1781a4485e]
aws_route_table.public_rt: Creating...
aws_route_table.private_rt: Creation complete after 5s [id=rth-Ba08A165e392b7d83]
aws_route_table_association.app_private_association: Creating...
aws_subnet.db_subnet: Creation complete after 4s [id=subnet-0292884b960cc928]
aws_route_table_association.db_private_association: Creating...
aws_route_table_association.app_private_association: Creation complete after 1s [id=rtbassoc-03324066d13d4f882]
aws_route_table_association.db_private_association: Creation complete after 1s [id=rtbassoc-0970b380f55fc9634]
aws_iam_instance_profile.ec2_profile: Creation complete after 9s [id=PBL-EC2-S3-Instance-Profile]
aws_route_table.public_rt: Creation complete after 2s [id=rtb-088ab4b919772c9ef]
aws_security_group.web_sg: Creation complete after 7s [id=sg-02d96807fd285eb35]
aws_security_group.observation_sg: Creation complete after 8s [id=sg-0ba07368f4618863d]
aws_security_group.app_sg: Creating...
aws_subnet.web_subnet: Still creating... [60m18s elapsed]
aws_security_group.app_sg: Creation complete after 3s [id=sg-044542e11ad3434a9]
aws_security_group.db_sg: Creating...
aws_security_group.db_sg: Creation complete after 3s [id=sg-0d2d668dc2cc2b8a1]
aws_subnet.web_subnet: Creation complete after 15s [id=subnet-09078535c393baafa]
aws_route_table_association.web_public_association: Creating...
aws_instance.pbl_web_server: Creating...
aws_route_table_association.web_public_association: Creation complete after 8s [id=rtbassoc-06a3da754b21403c]
aws_instance.pbl_web_server: Still creating... [60m10s elapsed]
aws_instance.pbl_web_server: Creation complete after 19s [id=i-0786446ff6eff6ff68c]

Apply complete! Resources: 29 added, 0 changed, 0 destroyed.
```

Figure 13: Terraform Deployment — 29 Resources.

Evidence 14: Current EC2 Instance

Place in: Unit 1 → EC2 Deployment

Status: Current evidence

What it proves: Shows the current EC2 instance i-0ac93d8d4f9f2f21b in the Web public subnet, running with the PBL-EC2-S3-Role and IMDSv2 required.

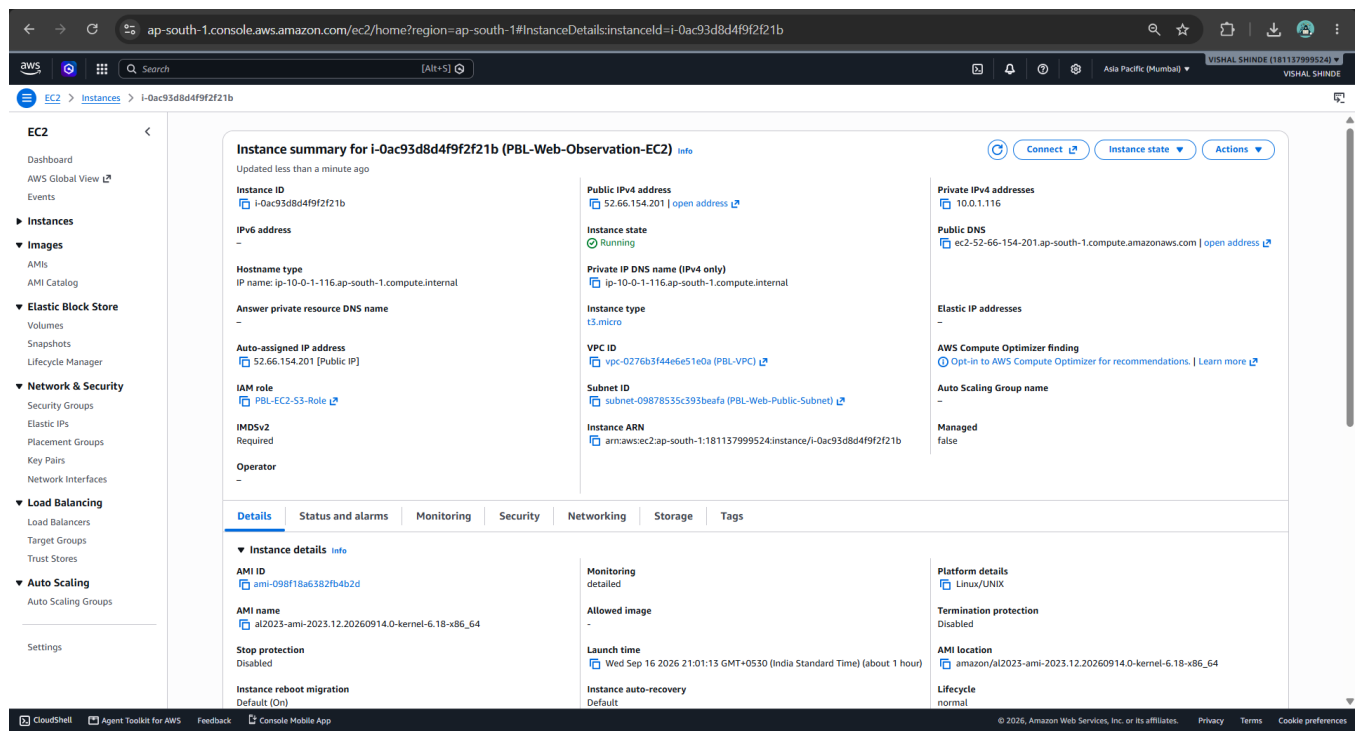


Figure 14: Current EC2 Instance.

Evidence 15: Systems Manager Session Manager Target

Place in: Unit 1 → Secure EC2 Administration

Status: Current evidence

What it proves: Shows the current EC2 node online and running in Session Manager.

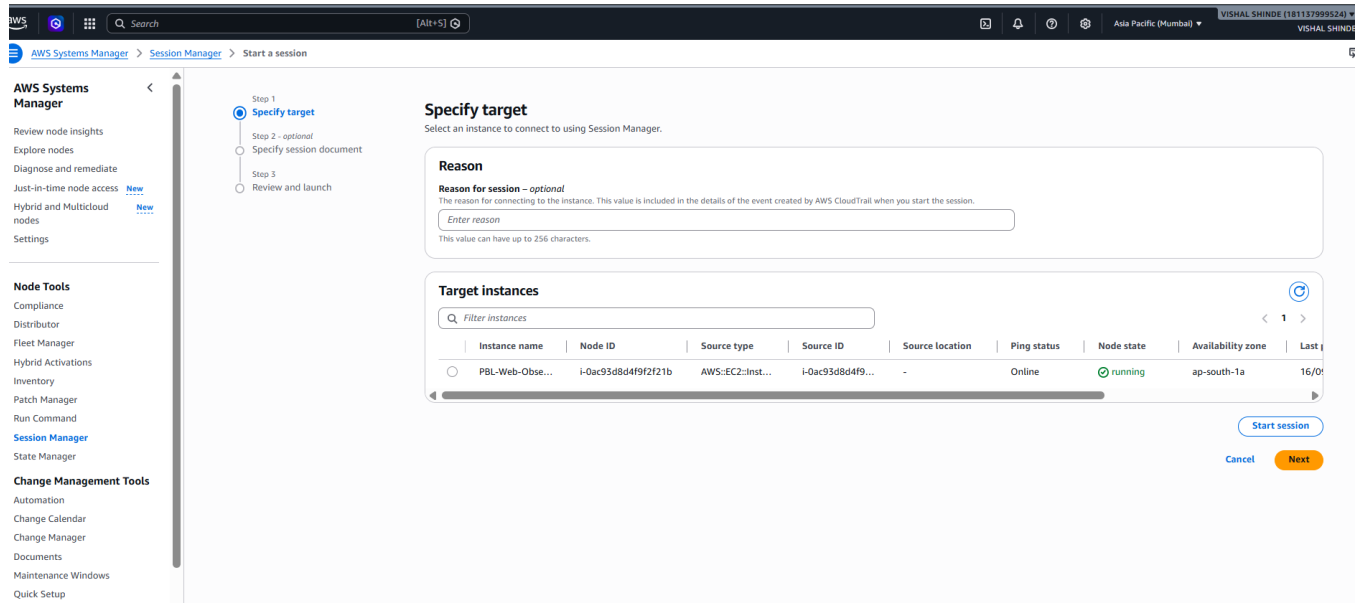


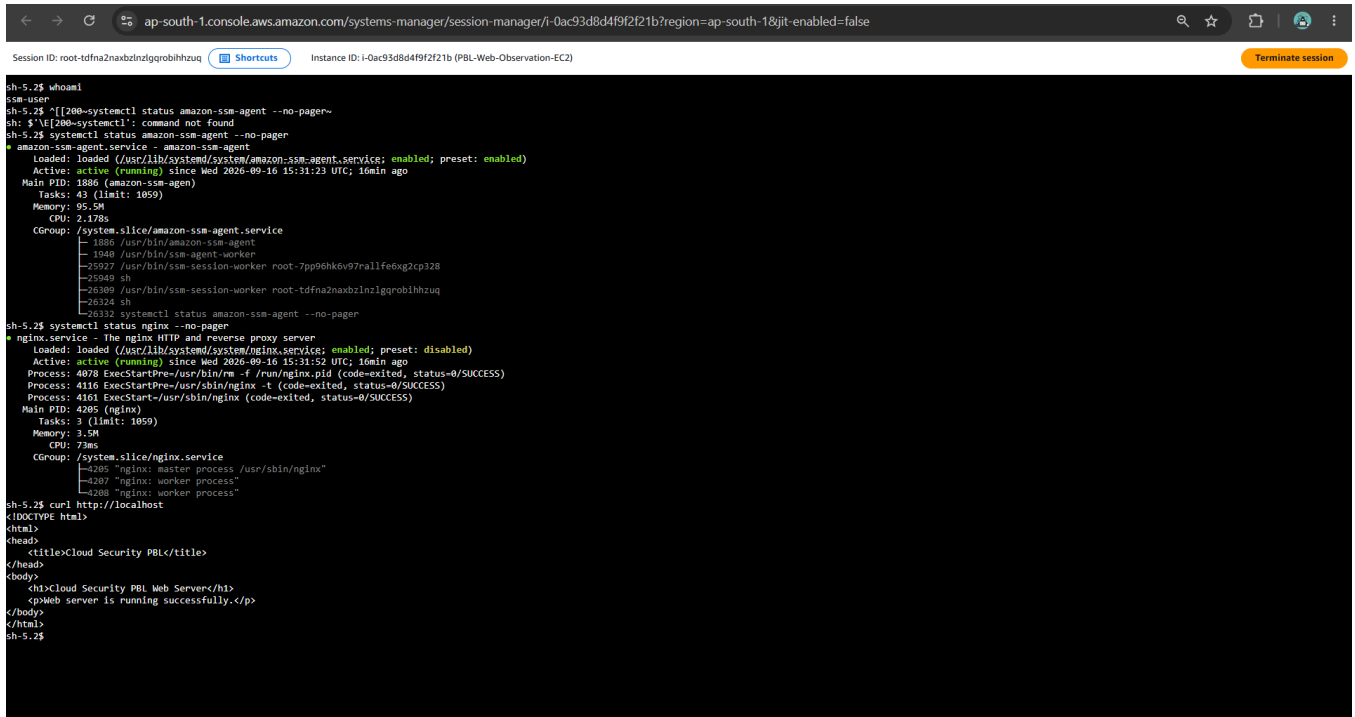
Figure 15: Systems Manager Session Manager Target.

Evidence 16: SSM, Nginx and Local Web Server Verification

Place in: Unit 1 → Web Server Verification

Status: Current evidence

What it proves: Shows the SSM agent active, Nginx active, and `curl http://localhost` returning the PBL web page.



```
ap-south-1.console.aws.amazon.com/systems-manager/session-manager/t-0ac93d8d4f9f2f21b?region=ap-south-1&jit-enabled=false
Session ID: root-tdfna2naxbzlnzlgqrobihhzuq Instance ID: i-0ac93d8d4f9f2f21b (PBL-Web-Observation-EC2)
sh-5.25 whoami
ssm-user
sh-5.25 systemctl status amazon-ssm-agent --no-pager~
sh: systemctl: command not found
sh-5.25 systemctl status amazon-ssm-agent --no-pager
amazon-ssm-agent.service - amazon-ssm-agent
Loaded: loaded (/usr/lib/systemd/system/amazon-ssm-agent.service; enabled; preset: enabled)
Active: active (running) since Wed 2026-09-16 15:31:23 UTC; 16min ago
Main PID: 1886 (amazon-ssm-agent)
Tasks: 43 (limit: 1059)
Memory: 95.5M
CPU: 2.178s
CGroup: /system.slice/amazon-ssm-agent.service
├─1886 /usr/bin/amazon-ssm-agent
├─1940 /usr/bin/ssm-agent-worker
├─25927 /usr/bin/ssm-session-worker root-7pp9hk6v97rallfe6xg2cp328
├─25949 sh
├─26309 /usr/bin/ssm-session-worker root-tdfna2naxbzlnzlgqrobihhzuq
├─26324 sh
└─26332 systemctl status amazon-ssm-agent --no-pager
sh-5.25 systemctl status nginx --no-pager
nginx.service - The nginx HTTP and reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
Active: active (running) since Wed 2026-09-16 15:31:52 UTC; 10min ago
Process: 4078 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
Process: 4116 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
Process: 4161 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
Main PID: 4205 (nginx)
Tasks: 3 (limit: 1059)
Memory: 3.5M
CPU: 72ms
CGroup: /system.slice/nginx.service
├─4205 "nginx: master process /usr/sbin/nginx"
├─4207 "nginx: worker process"
└─4208 "nginx: worker process"
sh-5.25 curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Cloud Security PBL</title>
</head>
<body>
<h1>Cloud Security PBL Web Servers</h1>
<p>Web server is running successfully.</p>
</body>
</html>
sh-5.25
```

Figure 16: SSM, Nginx and Local Web Server Verification.

Evidence 17: Current Web Server Browser Test

Place in: Unit 1 → Web Server Verification

Status: Current evidence

What it proves: Shows the deployed Cloud Security PBL web server successfully responding over HTTP.



Figure 17: Current Web Server Browser Test.

Evidence 18: Initial Observation EC2 Instance

Place in: Unit 1 → Internet-Facing Observation / Historical Evidence

Status: Historical — do not treat as current

What it proves: Shows the earlier observation EC2 instance with the previous public address. It is retained only as historical evidence because that instance was later replaced.

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

EC2 > Instances > i-0786446ff6efff60c

Instance summary for i-0786446ff6efff60c (PBL-Web-Observation-EC2)

Updated less than a minute ago

Instance ID
i-0786446ff6efff60c

IPv6 address
-

Hostname type
IP name: ip-10-0-1-146.ap-south-1.compute.internal

Answer private resource DNS name
-

Auto-assigned IP address
13.233.84.208 [Public IP]

IAM role
PBL-EC2-S3-Role

IMDSv2
Required

Operator
-

Public IPv4 address
13.233.84.208 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-10-0-1-146.ap-south-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-0276b3f44e6e51e0a (PBL-VPC)

Subnet ID
subnet-09878535c393beafa (PBL-Web-Public-Subnet)

Instance ARN
arnaws:ec2:ap-south-1:181137999524:instance/i-0786446ff6efff60c

Private IPv4 addresses
10.0.1.146

Public DNS
ec2-13-233-84-208.ap-south-1.compute.amazonaws.com | open address

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name
-

Managed
false

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance details

AMI ID
ami-0efe89759aa766662

AMI name
al2023-ami-minimal-2023.12.20260914.0-kernel-6.18-x86_64

Stop protection
Disabled

Instance reboot migration
Default (On)

Monitoring
detailed

Allowed image
-

Launch time
Wed Sep 16 2026 20:14:19 GMT+0530 (India Standard Time) (20 minutes)

Instance auto-recovery
Default

Platform details
Linux/UNIX

Termination protection
Disabled

AMI location
amazon/al2023-ami-minimal-2023.12.20260914.0-kernel-6.18-x86_64

Lifecycle
normal

Figure 18: Initial Observation EC2 Instance.

Evidence 19: Initial Observation Web Server

Place in: Unit 1 → Internet-Facing Observation / Historical Evidence

Status: Historical — do not treat as current

What it proves: Shows the earlier web server using the previous public address. Retained as historical evidence only.



Figure 19: Initial Observation Web Server.

Evidence 20: EC2 Replacement Plan

Place in: Unit 1 → Deployment Change / Historical Evidence

Status: Historical/change evidence

What it proves: Shows Terraform replacing the earlier EC2 instance. This explains why the old EC2 screenshots differ from the current instance.

```
# aws_instance.p01_web_server must be replaced
// resource "aws_instance" "p01_web_server" {
  # Warning! this attribute value will be marked as sensitive and will not
  # display in UI output after applying this change
  ami                         = (sensitive value) # Force replacement
  availability_zone           = "ap-south-1a" -> (known after apply)
  disable_api_termination    = false -> (known after apply)
  enable_efs_access_point     = false -> (known after apply)
  enable_primary_ipv6        = false -> null
  hibernation                 = false -> null
  host_id                    = (known after apply)
  host_resource_group_arn    = "arn:aws:ec2:ap-south-1:111179995254:instance/-4786446f6ff6dc" -> (known after apply)
  id                         = "i-0786446f6ff6dc" -> (known after apply)
  instance_initiated_shutdown_behavior = "stop" -> (known after apply)
  instance_lifecycle         = (known after apply)
  instance_state             = "running" -> (known after apply)
  ipv4_address_count         = 8 -> (known after apply)
  ipv4_addresses             = [] -> (known after apply)
  key_name                   = (known after apply)
  output_arn                 = (known after apply)
  password_data              = (known after apply)
  placement_group            = (known after apply)
  placement_group_id         = (known after apply)
  placement_partition_number = 0 -> (known after apply)
  primary_interface_id       = "eni-04d370b1c7c4d1e" -> (known after apply)
  private_dns                = "ip-18-bd-1-146-ap-south-1-compute.internal" -> (known after apply)
  private_ip                 = "18.91.146" -> (known after apply)
  public_dns                 = "ec2-13-233-84-206.ap-south-1.compute.amazonaws.com" -> (known after apply)
  public_ip                  = "13.233.84.206" -> (known after apply)
  secondary_private_ips      = [] -> (known after apply)
  security_groups            = [] -> (known after apply)
  spot_instance_request_id   = (known after apply)
  tags                       = {
    "Environment" = "Lab"
    "Name"        = "P01-Web-Observation-EC2"
    "Project"     = "Cloud-Security-P01"
    "Purpose"     = "Web and SSH Threat Observation"
  }
  tenancy = "default" -> (known after apply)
  user_data_base64 = (known after apply)
  # (13 unchanged attributes hidden)
  capacity_reservation_specification (known after apply) {
    capacity_reservation_preference = "open" -> null
  }
  cpu_options (known after apply) {
    core_count = 1 -> null
    threads_per_core = 2 -> null
    # (2 unchanged attributes hidden)
  }
  credit_specification {
    cpu_credits = "unlimited" -> null
  }
  ebs_block_device (known after apply)
  enclave_options (known after apply)
  enclave_options {
    enabled = false -> null
  }
  ephemeral_block_device (known after apply)
  instance_market_options (known after apply)
  maintenance_options (known after apply)
  maintenance_options {
    auto_recovery = "default" -> null
  }
  metadata_options (known after apply)
  metadata_options {
    http_endpoint = "enabled" -> null
    http_endpoint_ipv6 = "disabled" -> null
    http_put_response_hop_limit = 2 -> null
    http_tokens = "required" -> null
    instance_metadata_tags = "disabled" -> null
  }
}
```

Figure 20: EC2 Replacement Plan.

README — Submission Notes and Unit Mapping

Unit 1 contains: the AWS 3-tier network design (Web/App/DB), VPC and subnets, Internet Gateway and routing, Web/App/DB security groups, secure S3 configuration, Terraform deployment, EC2 deployment, Systems Manager access, and web-server verification.

Important: the older EC2 screenshots using the previous public address are historical. The current EC2 evidence is the instance ending in **...f21b**, shown in the current EC2 and Session Manager screenshots.

Pending requirement: the 24–48 hour SSH observation and failed-login log analysis must be supported by actual journal output before being marked complete. The screenshots in this report do not by themselves prove that requirement.

Submission note: Do not publish screenshots containing account identifiers, access-key material, or public IP addresses in public repositories or social media. Keep evidence in the intended academic submission channel.